

Wind Speed Forecasting Result Report

Season-wise Model Comparison, Selection, and Final Test Performance

1. Objective

The objective of this project is to forecast hourly wind speed using historical weather data. The target variable is WS, which represents wind speed. The forecasting task is a regression/time-series problem, so the results are evaluated using MAE, RMSE, MAPE, and MedAE. There are no confusion matrices because this is not a classification problem.

2. Pipeline Summary

- Load hourly weather data and create a clean datetime index.
- Fill missing values using forward and backward filling.
- Create cyclical time and wind-direction features.
- Split the data into four seasons.
- Split each season chronologically into train, validation, and test sets.
- Tune candidate models using only training data.
- Compare all candidate models on validation data.
- Select the best model per season using the lowest mean validation RMSE.
- Evaluate the selected model on the unseen test set.
- Generate probabilistic intervals, explainability outputs, saved models, and final 24-hour forecasts.

3. Data Split

Each season was split chronologically to avoid future data leakage. The split ratio was 70% training, 15% validation, and 15% testing.

Season	Train	Validation	Test
Spring	1481	317	318
Summer	1481	317	318
Autumn	1465	314	314
Winter	1465	314	314

4. Models Evaluated

The following models were evaluated. Model D is mainly used for prediction intervals rather than ordinary point-forecast comparison.

Model	Description
A	Recursive LightGBM
B	Direct LightGBM
C	Baseline equivalent-date model
D	Quantile LightGBM interval model
E	Recursive XGBoost
F	Recursive Ridge

Model	Description
G	Recursive Random Forest
H	Stacking ensemble using A, E, and F

5. Model Selection Rule

For each season, all candidate models were compared on the validation set. The model with the lowest average validation RMSE across horizons 1, 2, 3, and 24 was selected as the best model. The test set was not used for model selection; it was used only for final evaluation after the best model had already been selected.

6. Best Model Per Season

Season	Best Model	Model Name	Validation RMSE
Spring	G	Recursive Random Forest	2.3289
Summer	B	Direct LightGBM	1.0502
Autumn	H	Stacking Ensemble	1.5358
Winter	H	Stacking Ensemble	1.6751

7. Validation Comparison Matrix

The following tables show the mean validation RMSE ranking for all models in each season. Lower RMSE is better.

Spring

Rank	Model	Mean Validation RMSE
1	G	2.3289
2	A	2.3518
3	E	2.4365
4	F	2.5331
5	B	2.5805
6	H	3.6210
7	C	4.8938

Summer

Rank	Model	Mean Validation RMSE
1	B	1.0502
2	C	1.1633
3	G	1.1854
4	H	1.2226
5	A	1.2417
6	F	1.3910
7	E	1.4143

Autumn

Rank	Model	Mean Validation RMSE
1	H	1.5358
2	E	1.6009
3	B	1.6192
4	A	1.7502
5	G	1.8198
6	F	2.0846
7	C	3.6433

Winter

Rank	Model	Mean Validation RMSE
1	H	1.6751
2	G	1.8026
3	A	1.9824
4	B	2.0080
5	F	2.4404
6	E	2.5411
7	C	3.5936

8. Final Test Results

After selecting the best model using validation results, the selected model was evaluated on unseen test data. The following tables show final test performance at forecast horizons 1, 2, 3, and 24 hours.

Spring - Best Model G

Horizon	MAE	RMSE	MAPE	MedAE
1	0.8058	1.0360	0.1925	0.7905
2	0.8923	1.1660	0.2476	0.8874
3	1.6695	2.6510	0.3875	1.1688
24	1.7973	2.6302	0.4616	0.9783

Summer - Best Model B

Horizon	MAE	RMSE	MAPE	MedAE
1	0.4475	0.4955	0.1984	0.4728
2	0.5498	0.7990	0.2337	0.3540
3	0.6754	0.9916	0.2270	0.4358
24	0.4114	0.4599	0.1859	0.3992

Autumn - Best Model H

Horizon	MAE	RMSE	MAPE	MedAE
1	1.3404	1.4364	0.6570	1.2158
2	1.3050	1.5396	0.5476	1.3001
3	1.9761	2.3692	0.6239	1.4874
24	1.1912	1.3760	0.4778	1.3085

Winter - Best Model H

Horizon	MAE	RMSE	MAPE	MedAE
1	4.6769	5.4413	0.5606	4.4208
2	4.6937	5.4636	0.6400	4.5417
3	4.5150	5.4403	0.5600	4.4914
24	4.5732	5.3818	0.5616	4.9804

9. Probabilistic Forecast Interval Results

Prediction intervals were generated using bootstrap, conformal, and quantile methods. Coverage means the proportion of actual values captured inside the predicted interval. Average width measures how wide the intervals are. Higher coverage is better, but very wide intervals are less precise.

Spring

Method	Coverage	Average Width
Bootstrap	0.4583	4.1835
Conformal	0.8750	8.9250
Quantile	0.5833	6.7477

Summer

Method	Coverage	Average Width
Bootstrap	0.5000	2.2311
Conformal	0.6250	3.9101
Quantile	0.8750	11.6887

Autumn

Method	Coverage	Average Width
Bootstrap	0.9583	5.7690
Conformal	1.0000	6.7674
Quantile	0.9583	15.4308

Winter

Method	Coverage	Average Width
Bootstrap	0.4583	2.6896
Conformal	1.0000	10.6682
Quantile	1.0000	27.7308

10. Interpretation of Results

- The best model was not the same for every season, showing that wind behavior changes across the year.
- Spring performed best with Recursive Random Forest.
- Summer performed best with Direct LightGBM and had the strongest final test performance.
- Autumn and winter performed best with the stacking ensemble.
- Winter test performance was weaker than the other seasons, suggesting the winter test period was more difficult or had different wind behavior compared with validation.
- The validation comparison justifies model selection, while the test results show final unseen-data performance.

11. Final Conclusion

The seasonal forecasting approach was successful because it allowed a different best model to be selected for each season. Model selection was performed correctly using validation RMSE, and the final test set was used only for unbiased performance reporting. The final selected models were G for spring, B for summer, H for autumn, and H for winter. Overall, the project demonstrates a complete forecasting workflow including preprocessing, feature engineering, model tuning, validation comparison, best-model selection, final testing, probabilistic intervals, explainability, and production forecasting.